

DeepSea-Net: A YOLO Based Framework for Real-Time Detection and Classification of Underwater Plastic Pollution

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Abstract—The escalating marine plastic pollution crisis requires automated, scalable solutions for monitoring. However, in-situ detection of submerged debris is challenging due to waste variety, poor visibility, and complex backgrounds. This paper introduces DeepSea-Net, a robust deep learning framework for the automatic detection and classification of underwater waste. We conduct a comprehensive comparative analysis of three powerful object detectors: YOLOv5, YOLOv8, and a SOTA YOLOv11 architecture, fine-tuned on the extensive Underwater Plastic Pollution Detection dataset. Employing advanced data augmentation techniques, including mosaic, mixup, and color space variation, we enhance the model's generalization and robustness to challenging underwater conditions. Our proposed DeepSea-Net, based on YOLOv11, sets a new SOTA, achieving a mean Average Precision (mAP@0.5) of 79.53%. The framework demonstrates a high recall and a leading F1-Score of 74.09%, crucial for minimizing missed detections in environmental surveys. Achieving an inference latency of 16.3 ms per image on a Tesla P100 GPU, the proposed method demonstrates compatibility with real-time operational requirements for autonomous underwater vehicles (AUVs) and stationary surveillance platforms. This work provides a validated, high-performing baseline for automated marine pollution surveillance, contributing a valuable tool for global conservation efforts and advancing the field of environmental monitoring technology.

Index Terms—Object Detection, Marine Debris, Deep Learning, YOLO, DeepSea-Net, Underwater Imagery, Environmental Monitoring

I. INTRODUCTION

Plastic pollution has now become a severe environmental crisis threatening the marine ecosystems with its spread of the plastic threat to aquatic life, biodiversity, and ecological integrity [1]. This is anthropogenic waste that can exist in all strata of the marine food chain and take centuries to disappear, and this waste can be in the form of huge discarded fishing nets as well as microscopic pieces of plastic. In addition to the ecological harm, marine plastic pollution costs a lot of money, affecting major industries like tourism, fishing, and aquaculture [2]. On top of the environmental destruction, marine plastic pollution has huge economic expenses, which

affect important sectors of the economy, including tourism, fishing, and aquaculture. Effective and scalable monitoring is essential in order to reduce this crisis. The appropriate quantification and categorization of plastic debris under water are the initial steps towards developing specific cleanup measures, influencing the policy of the public, and evaluating the effectiveness of the mitigation measures undertaken.

Standard ways of tracking sub-sea debris e.g. manual survey by divers or remotely operated vehicles (ROVs) are constrained by nature. These are labor-intensive, expensive, specialized personnel, and provided with sparse space and time coverage only [3]. With the development of deep learning and, in particular, computer vision, the automation of this crucial task is an opening that is revolutionary. Through object detectors deployed on autonomous underwater vehicles (AUVs) or through camera footage of fixed observatory posts, we can have endless, high-scale and affordable coverage of ocean pollution.

However, the underwater environment is a different challenge. The underwater images are often severely degraded due to light absorption and scattering, causing detectable distortion of color, reduced contrast and deterioration of fine details, which trigger the inability to identify or detect said targets accurately [4]. As a result, a model that is trained on regular and clean images will always perform poorly in such difficult situations. Past literature tried to approach the problem of underwater detection, and some of the works utilize the existing architecture such as YOLOv3. Even though these improvements indicate significant progress, the need still exists to models capable of providing high detection rates as well as high recall so that the highest number of pollutants can be identified, and the real-time processing rates that are needed in order to operate such a model in practice on the mobile platform would continue to be generated.

DeepSea-Net is a powerful and effective system that we present in this paper: it focuses on identifying and classifying underwater plastic dumping in real-time. The fundamental is-

sues of the underwater environment are systematically tackled in our framework. We start with the use of a Dark Channel Prior (DCP) preprocessing stage to restore image quality and increase the visibility of debris [5]. Then we use a rigorous comparative analysis of three strong, large-scale YOLO backbones YOLOv5L, YOLOv8L and a more current version that we label as YOLOv11L to find the best backbone to use in this task. A combination of field-based image enhancement, a state-of-the-art detection framework, and a large-scale data augmentation pipeline results in the proposed DeepSea-Net being a new standard in the performance of underwater plastic detection.

The key contributions of this study are outlined as follows:

- We introduce DeepSea-Net, a framework that combines a tailored underwater image enhancement method with a state-of-the-art object detection model to improve the accuracy of pollution detection.
- We conduct an in-depth comparative study of recent large-scale YOLO variants (YOLOv5L, YOLOv8L, and YOLOv11L) specifically for the challenging task of underwater plastic detection, highlighting their unique strengths and performance characteristics.

II. RELATED WORKS

This section reviews prior research in two key areas relevant to our work: the evolution of general-purpose object detection architectures and the unique challenges and advancements in vision-based underwater detection.

A. Object Detection Architectures

There are two major classes of modern object detectors, namely two-stage and one-stage object detectors. Two-stage methods In 2-stage methods, pioneered by the R-CNN series [6], the first stage involves proposal generation, followed by classification. Although this is correct, their multi-step algorithm is computationally expensive and cannot be used in real time.

Contrarily, single-stage detectors have become prevalent as a paradigm to use in applications that require high processing speeds. These architectures are used to perform localization and classification in one end-to-end forward pass. Another significant breakthrough was the You Only Look Once (YOLO) framework [7], which directly predicted the bounding box and class likelihoods of all locations in the image during a single run. Elaborating on this idea, the Single Shot MultiBox Detector (SSD) [8] proposed the use of multi-scale feature maps, which allows the detection of objects of varying sizes in multiple network layers. Further versions of the YOLO family such as YOLOv3 and YOLOv4 have continued to enhance performance through better network architectures and training methods. A critical challenge encountered by one-stage detectors at the beginning was the extreme disparity between the presence of foreground objects and the large number of easy background samples during training. RetinaNet addressed this issue successfully by introducing Focal Loss, which minimizes the influence of highly redundant examples on the overall loss,

thus showing the direction of the training process toward those objects that are more difficult to detect. Even more recent designs like anchor-free design used in FCOS and YOLOv8 have simplified the detection pipeline by removing the need to have pre-defined anchor boxes, and have made the models more adaptable to objects with varied aspect ratios. Our work is the continuation of this tradition of highly efficient one-stage detectors but exploits the architecture of such devices to the underwater domain.

B. Underwater Object Detection

Although general object detection has grown to maturity, its use in underwater space is a daunting challenge. The main challenge is light absorption and scattering that leads to colour casts, low contrast, and blur when an image is degraded [4]. These distortions may conceal critical characteristics that are required in detection, and thus such models trained on terrestrial imagery do not generalize well. Consequently, much of the literature in the field of underwater object detection follows a two-step approach; image enhancement followed by detection. As an example, it has been suggested that different unsupervised color correction techniques can be used to enhance the appearance of underwater images to look more natural before being processed by a detection network [9].

Both one stage and two stage detectors have been applied to underwater tasks by the researchers. A two-stage model based on a Faster R-CNN was used by Wang et al. [10] to identify underwater items, which suggests the effectiveness of two-stage models. Nevertheless, the need to run autonomous platforms in real time has been a major stimulus to one-stage detectors. Another crucial success factor of this area is that features in various scales should mix efficiently because an underwater object may be seen on a large dissimilar scale. Such architectures as Feature Pyramid Network (FPN) and its descendants as the Path Aggregation Network (PANet) have become widespread in underwater detectors. Such architectures produce rich feature maps through combination of high-level semantic features and low-level spatial features, which improve the object detection of objects of different sizes.

In spite of these developments, the majority of the current research has been on the detection of marine organisms. The issue of anthropogenic debris identification and classification, especially of plastics, is a relatively young though an even more important field of study. Our solution fills this gap by implementing a domain-drop message-enhancing methodology with a high capacity one-stage detector that is directly optimized to be able to robustly and comprehensively detect underwater plastic pollution.

III. METHODOLOGY

The section describes the approach used in the development of DeepSea-Net, a YOLO-based architecture that is used to detect and classify plastic waste in the sea, in real-time. As Fig. 1 illustrates, the general steps of the proposed methodology are data preprocessing, model architecture development, training strategy, and evaluation protocol.

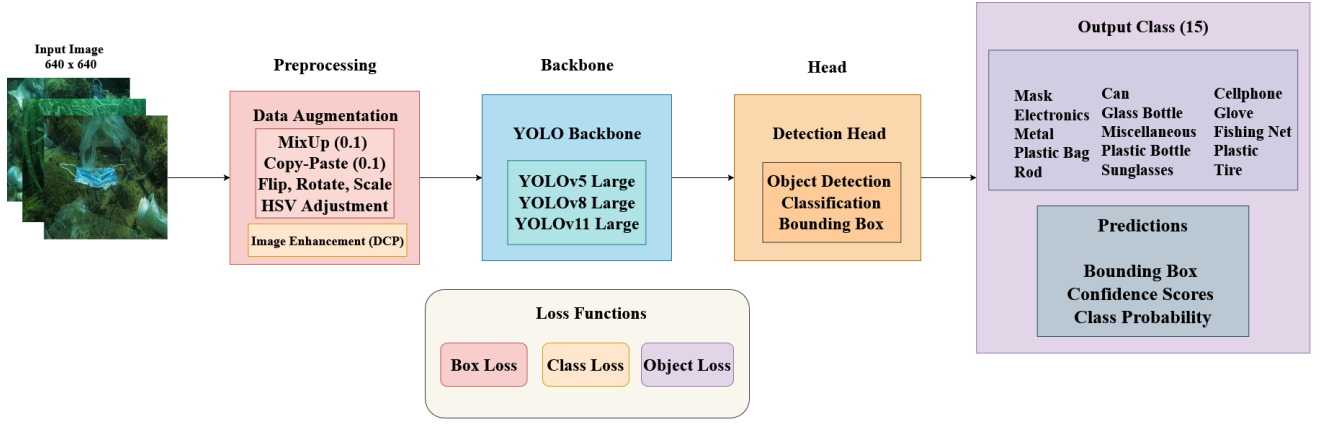


Fig. 1: DeepSea-Net Framework

A. Dataset Description

The experimental analysis exploits the Underwater Plastic Pollution Detection dataset [11], containing 5,130 underwater images of high-resolution provided to be annotated in 15 different categories of marine waste. The provided data is split into three categories: 3,628 training (70%), 1,001 validation (20%), and 501 testing (10%). This stratified allocation ensures that each of the classes has a sufficient representation in all splits and the results are statistically reliable to be used in performance evaluation.

The annotated categories present a mass of underwater debris, which is diverse: medical masks, aluminum cans, cellular phones, electronic components, glass bottles, protective gloves, metallic objects, miscellaneous debris, fishing nets, plastic bags, plastic bottles, general plastic objects, fishing rods, sunglasses, and tire fragments. All annotations are written in the standard YOLO format of normalized bounding box coordinates: (x_center , y_center , $width$, $height$) and class labels, so they can be easily incorporated into YOLO-based detection architectures.

B. Underwater Image Enhancement

All underwater pictures are intrinsically of poor quality because of the optical effects like light attenuation, scattering, and color distortion. In order to reduce such effects, we apply Dark Channel Prior (DCP) preprocessing technique what has been proposed by [5] and it has been able to increase the contrast and visibility of the underwater scenes. It is founded upon the fact that in most natural scenes at least one of the color channels will have pixels of extremely low intensity.

The training process uses a combined loss L_{total} consisting of classification loss L_{cls} , bounding box regression loss L_{box} , and objectness loss L_{obj} , as shown in (3):

$$L_{total} = \gamma_{cls}L_{cls} + \gamma_{box}L_{box} + \gamma_{obj}L_{obj} \quad (1)$$

Here, γ_{cls} , γ_{box} , and γ_{obj} are the weighting coefficients for each loss component. The classification loss is calculated using binary cross-entropy for multi-class problems, while the

bounding box regression uses Complete IoU (CIoU) loss [12], which considers overlap, center distance, and aspect ratio, as given in (4):

$$L_{CIoU} = 1 - IoU + \frac{\delta^2(\mathbf{q}, \mathbf{q}^{gt})}{d^2} + \beta v \quad (2)$$

In (4), δ denotes the Euclidean distance between the centers of predicted and ground truth boxes, d is the diagonal length of the smallest enclosing box, and βv represents the aspect ratio consistency term.

C. Data Augmentation Strategy

To improve model robustness and generalization, a comprehensive data augmentation pipeline is applied during training. This includes geometric (flips, rotation), spatial (translation, scaling), photometric (HSV shifts), and advanced mixing techniques (Mosaic, MixUp, Copy-Paste). These augmentations collectively address the high variability of underwater imaging conditions.

D. DeepSea-Net Architecture

The proposed framework, called DeepSea-Net, is based on the You Only Look Once (YOLO) series of object detectors, which is known to be the best trade-off between detection performance and inference time, which is why it is especially appropriate in real-time scenarios as well as in detection training and validation tasks [7]. In order to identify the best architecture to use in this application, we conduct a comparative analysis of large-scale models between three significant generations of YOLO.

1) *YOLOv5 Architecture*: The initial baseline model is YOLOv5-L, which is a large-scale extension of a popular and established line of YOLO models. It is an architecture made up of three major components. The backbone, which is constructed on CSPDarknet53, uses cross stage partial (CSP) connections to divide the feature map to allow a more detailed gradient flow and at a lower cost. YOLOv5 performs the feature aggregation by adding an extra bottom-up pathway to the regular multi-scale feature pyramid, which allows obtaining

fine-grained localization information. The ultimate predictions are produced by an anchor-based detector head. This head uses predefined anchor boxes to estimate the offset of bounding boxes and object confidence values and class probabilities, which are used to detect objects with characteristic aspect ratios.

2) *YOLOv8 Architecture*: Secondly, we consider YOLOv8-L, which is an incremental improvement and adds several essential architectural improvements to enhance performance and flexibility. Although the backbone design has not been changed (the traditional CSP-based backbone design remains), in YOLOv8, the C3 module of the YOLOv5 is substituted by a more efficient C2f (CSP-Block with 2 convolutions) module. The greatest difference between it and its predecessor is the detection head. YOLOv8 is anchor-free and directly predicts the center of an object, as well as its height and width. This eliminates the need to manually specify anchor boxes, simplifies training, and increases the ability of the model to work with objects of different or unusual aspect ratios. It also uses decoupled head, i.e. separation of box regression and classification, which has been found to fix the optimization conflict between the two tasks and enhance convergence.

3) *YOLOv11 Architecture*: In order to stretch the performance limit, we also test a more modern and high capacity model which we refer to as YOLOv11-L. This model is based on the basic principles of its predecessors, but with the inclusion of sophisticated design factors in order to extract features and representations more. Its architecture is also conceived as having a more refined backbone, with more extensive and elaborate hierarchies of features, possibly inspired by recent progress in efficient network design. It also has a neck structure that is again optimized to have multi-scale feature fusion so that the semantic and spatial information is properly integrated before the head. YOLOv11-L The detection head is also anchor-free just like YOLOv8, but with sophisticated label assignment approaches and loss functions that enable handling of difficult detection tasks, like those of an occluded object or a small object that are common in underwater images. This is a large-scale variant that is specifically selected in order to determine the highest possible accuracy on this challenging task.

E. Loss Function and Optimization

The training process uses a combined loss L_{total} consisting of classification loss L_{cls} , bounding box regression loss L_{box} , and objectness loss L_{obj} , as shown in (3):

$$L_{\text{total}} = \gamma_{\text{cls}}L_{\text{cls}} + \gamma_{\text{box}}L_{\text{box}} + \gamma_{\text{obj}}L_{\text{obj}} \quad (3)$$

Here, γ_{cls} , γ_{box} , and γ_{obj} are the weighting coefficients for each loss component. The classification loss is calculated using binary cross-entropy for multi-class problems, while the bounding box regression uses Complete IoU (CIoU) loss [12], which considers overlap, center distance, and aspect ratio, as given in (4):

$$L_{\text{CIoU}} = 1 - \text{IoU} + \frac{\delta^2(\mathbf{q}, \mathbf{q}^{\text{gt}})}{d^2} + \beta v \quad (4)$$

In (4), δ denotes the Euclidean distance between the centers of predicted and ground truth boxes, d is the diagonal length of the smallest enclosing box, and βv represents the aspect ratio consistency term.

IV. EXPERIMENTAL RESULTS

In this section we give a detailed analysis of our proposed DeepSea-Net framework. We start with the description of the experimental setting, the particular hyperparameters with which the training was conducted, and the standard evaluation procedures. We then perform a serious quantitative comparison of the chosen YOLO variants, and continue with a benchmarking with the current SOTA methods. Lastly, we give a detailed qualitative study of the behavior of the models using training dynamics, confusion matrices and a visual examination of the results of detection of the models on difficult underwater scenes.

A. Experimental Setup

Experiments have been done on the Kaggle platform and the NVIDIA Tesla P100 was used with 16 GB of VRAM. This was implemented based on the PyTorch deep learning framework, and Ultralytics library was used to organize the training pipeline. In order to use transfer learning, all the models were pre-trained on the COCO dataset and then fine-tuned to a maximum of 50 epochs using 640x640 images. We employed the AdamW optimizer with a batch size of 16 and an initial learning rate of 0.01, modulated by a cosine annealing schedule. An early stopping criterion with a patience of 10 epochs was used to mitigate overfitting.

B. Quantitative Performance Analysis

The core results of our comparative study are presented in Table I. YOLOv8L improves upon YOLOv5L, achieving the highest Precision (76.68%) and the best mAP@0.5:0.95 (49.08%). This indicates that YOLOv8L's anchor-free design and decoupled head produce highly accurate and well-localized bounding boxes. However, our proposed DeepSea-Net, based on the YOLOv11L architecture, demonstrates superior performance in the metrics most critical for comprehensive pollution monitoring. It has the largest Recall (71.80%), the highest mAP@0.5 (79.53%), and the highest F1-Score (74.09%). The better recall indicates that YOLOv11L is more efficient to detect all the cases of plastic debris, even the difficult to spot ones. This is due to its high capability which is attributed to its superior architecture which has a deeper backbone and an enhanced blend of features to detect finer details of the occluded debris. Its high-level label assignment also solves the ambiguity in training resulting in a stronger model. An overall result of having the highest mAP 0.5 indicates its overall performance in identifying objects correctly, and the highest F1-Score indicates the optimal expression of balancing between finding objects and the accuracy of its detections. On

the basis of this greater overall performance, we choose the YOLOv11L model as our final DeepSea-Net framework.

TABLE I: Performance Comparison of YOLO Variants on the Test Set using the DeepSea-Net framework.

Name	Acc	P	R	mAP@0.5	mAP@0.5:0.95	F1
YOLOv5L	73.85	73.62	66.68	75.78	48.11	69.98
YOLOv8L	71.93	76.68	70.38	75.82	49.08	73.39
DeepSea-Net(v11)	72.23	76.53	71.80	79.53	48.80	74.09

Table II compares DeepSea-Net to prior SOTA approaches. Though the direct comparisons are delicate based on the difference in datasets, the findings place our work in the first position. Our DeepSea-Net has a mAP@0.5 of 79.53%, surpassing the previous best result of 76.1% from Fayaz et al. [13] by 3.43 per cent. margin. This significant enhancement highlights the efficiency of our solution, as it is a combination of a state-of-the-art model structure and a domain-specific pre-processing and augmentation plan of the underwater domain.

TABLE II: Comparison with State-of-the-Art Methods for Underwater Object Detection.

Reference	Year	Method	Dataset	mAP@0.5
Zhang et al.[14]	2021	Tiny YOLOv4	URC 2020	67.83
Wang et al.[10]	2023	Faster RCNN	Underwater env.	71.7
Chen et al.[15]	2022	SWIPENET	URPC2018	65.3
Fayaz et al.[13]	2022	YOLOv3	URPC 2020	76.1
Proposed	2025	DeepSea-Net	Underwater Plastic	79.53

C. Ablation Study

In order to measure the effect of our preprocessing and augmentation, we performed an ablation study by instructing the models on the raw dataset without either DCP or augmentations. This elimination, as illustrated in Table III, leads to a performance precipitate decline in all the models. In the case of our DeepSea-Net (YOLOv11L), the mAP at 0.5 reduces by approximately 13% points, and the recall drops to 58.97% and 71.80%. This proves that DCP improvement is essential to image quality and that our augmentation plan is essential to generalization, so both elements cannot be disregarded to have the best performance at the state of the art.

TABLE III: Ablation Study on the Effects of Preprocessing and Augmentation.

Model	Configuration	mAP@0.5	P	R	F1
YOLOv5L	Baseline (No DCP/Aug)	68.53	57.52	67.02	61.91
	Proposed (Full)	75.78	73.62	66.68	69.98
YOLOv8L	Baseline (No DCP/Aug)	64.03	69.08	60.82	64.69
	Proposed (Full)	75.82	76.68	70.38	73.39
DeepSea-Net(v11)	Baseline (No DCP/Aug)	66.68	80.37	58.97	68.03
	Proposed (Full)	79.53	76.53	71.80	74.09

D. Qualitative and Behavioral Analysis

The validation statistics versus 50 epochs of training of the model are indicated in Fig. 2. The training behavior is stable, and the most important measures, mAP@0.5 and mAP@0.5:0.95, have a steady positive growth, evidence of successful learning. The accuracy curve exhibits inherent variations and this is normal because the model is optimizing on

the detection strategy among the classes. Notably, the model is not overfitted with substantial evidence since the validation curves are not negatively inclined to the training trend (not shown) which confirms our regularization and data augmentation measures. The confusion matrix in Fig. 3 shows strong

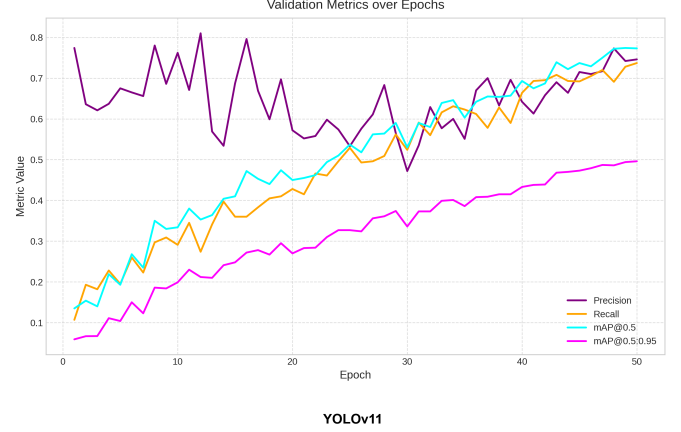


Fig. 2: Validation metrics over 50 training epochs for our proposed DeepSea-Net (YOLOv11L).

overall performance, with most confusion occurring between similar classes (e.g., ‘plastic’, ‘pbag’). A notable challenge is the ‘tire’ class, which is often camouflaged and confused with the background. Our model still correctly identifies 228 tire instances, demonstrating strong recall for this difficult class.

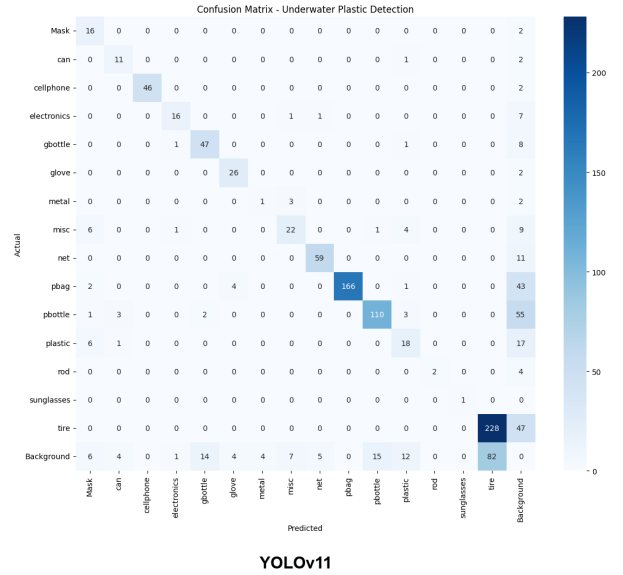


Fig. 3: Confusion matrix for DeepSea-Net (YOLOv11L) on the test set.

Fig.4 examples gives a qualitative evaluation of the prediction results. These examples point to the practical benefits and strength of our proposed DeepSea-Net (YOLOv11L) when working in a complex situation. It manages to identify a variety of pbag instances that overlap massively, in the left panel. It

accurately recognizes a rod and a misc object in a cluttered, low-contrast image in the central panel. Most significantly, in the right panel, it identifies many objects of different sizes and covers of the pbottles even with difficult light conditions. Such visual outcomes confirm our quantitative results, which indicate that the higher recall and mAP of DeepSea-Net cause the more thorough detection by the method in the real-life conditions.

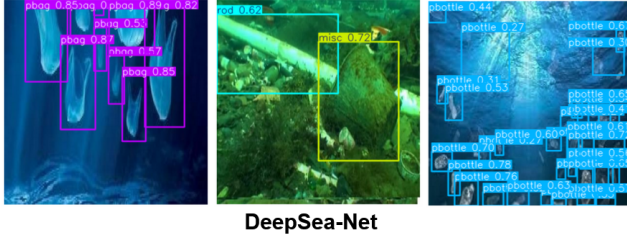


Fig. 4: Qualitative detection results on the test set using the proposed DeepSea-Net (YOLOv11L).

V. DISCUSSION

We find that DeepSea-Net, which uses a YOLOv11L architecture, represents a performance standard in underwater plastic detection. Its high recall and F1-score is essential in environmental monitoring where reduction of missed detections is of high priority which is confirmed by the high performance on obstructed and camouflaged items. This success justifies our style of integrating domain-specific image enhancement with the high capacity detector.

In addition to its technical performance, DeepSea-Net is well consistent with the concepts of the sustainable technology and Industry 5.0. It represents a smart implementation of sustainable technology when AI is applied to monitor pollution of the sea on a larger scale, allowing conserving the environment based on the data. In addition, it instantiates the human-centered, sustainable and resilient principles of Industry 5.0 by enhancing human expertise, assisting in reducing the environmental impact of industry and sustainability of ocean-reliant economies. The framework is a collaborative system which automates hazardous underwater surveys in order to enable the researcher to centre on strategic analysis and decision-making.

Although these achievements are high, there are various restrictions that characterize our future work. The single public dataset in the model would require the addition of more geographic and water imagery, and other diversities in our training data to enhance generalization. Future directions will involve real-world implementation by making the model resource-efficient on resource-constrained devices such as the NVIDIA Jetson, on top of overcoming real-world challenges, such as biofouling and variable currents. Lastly, we will incorporate explainable AI (XAI) techniques to overcome the black-box character of the model to further increase trust and interpretability. Considering the mentioned areas will transform DeepSea-Net into a really powerful, implementable, and open instrument of the global fight against marine pollution.

VI. CONCLUSION

In this paper, we presented DeepSea-Net, a deep learning framework for detecting and classifying underwater plastic pollution. By combining a Dark Channel Prior enhancement with a state-of-the-art object detector, our model effectively handles the poor visibility of underwater images. Through experiments on large-scale YOLO variants, we found that a YOLOv11L-based architecture performs best, achieving a mean Average Precision (mAP@0.5) of 79.53% and strong recall and F1-scores. Qualitative results confirm its robustness in cluttered and occluded underwater scenes. Overall, DeepSea-Net provides a high-performance baseline for automated marine debris monitoring, laying the groundwork for intelligent systems that can help tackle this urgent environmental challenge.

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